

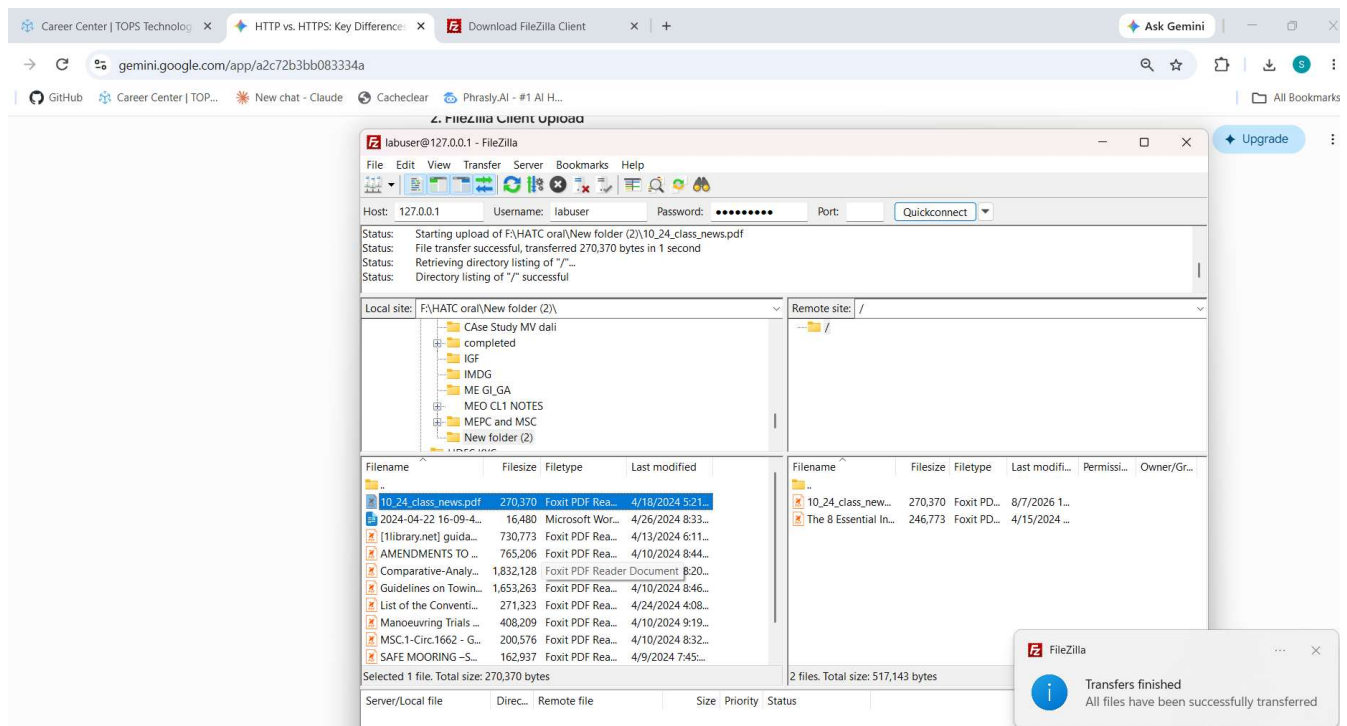
Day 4. Other Protocols in Action

Task 1. Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.

Download and run the FileZilla Server setup on Windows.

- Open FileZilla Server Administration.
- Go to Server - Configure - Users.
- Click Add to create a user account (e.g., Username: labuser, set a password under Credentials).
- Go to Mount points - Click Add:
- Virtual path: /
- Native path: C:\ftp (or any local directory on your PC).
- Grant Read and Write permissions.
- Click OK to save and start the server.

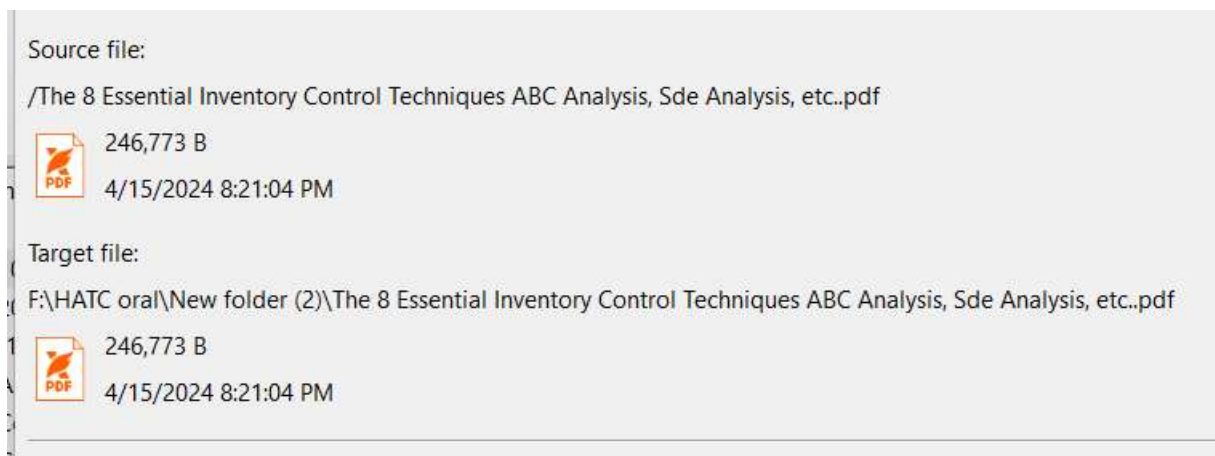
Task 2. Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.



Task 3. Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.

In FileZilla Client, navigate to a different local folder in the **Local site** pane (e.g., C:\Downloads).

- In the **Remote site** pane, right-click test_image.png $\rightarrow$ **Download**.
- **Verification:**
- Open File Explorer, right-click both the original and downloaded images, and select **Properties**.
- Compare the **Size** and **Size on disk** (in bytes) to ensure exact file parity.



Task 4. Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites). Hint: Think about how network admins monitor device health, bandwidth, or uptime.

Real-World SNMP Use Cases

- **ISP Bandwidth & Traffic Monitoring:** ISPs use SNMP polling on core routers to track bandwidth utilization across interfaces via OID metrics. This helps identify network congestion, plan capacity, and enforce Data Usage Limits/SLA commitments.
- **Campus Wi-Fi Infrastructure Health:** University IT departments use SNMP traps to monitor hundreds of Access Points (APs). If an AP drops offline, experiences high packet loss, or exceeds CPU/memory thresholds, an SNMP alert is sent immediately to the Network Operations Center (NOC).
- **Data Center Server Uptime & Environment Monitoring:** Large web services monitor physical server health (power supply status, fan speeds, hardware temperatures) and link status using SNMP agents integrated into Baseboard Management Controllers (iDRAC/iLO).

Task 5. Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.

Feature	FTP (File Transfer Protocol)	HTTP (Hypertext Transfer Protocol)
Port Number	Ports 20 (Data) and 21 (Control)	Port 80 (HTTP) / Port 443 (HTTPS)
Security	Insecure by default (plain text credentials); requires FTPS or SFTP for encryption.	Insecure on Port 80; uses TLS/SSL encryption by default via HTTPS on Port 443.
Data Transfer Method	Two-connection architecture: Uses separate control and data channels.	Single-connection architecture: Control headers and payload transfer over the same TCP session.
Primary Use Cases	Bulk file management, remote directory maintenance, uploading website builds.	Rendering web pages, API communication, web browsing, downloading web assets.
Authentication	Built-in native user authentication required by default (supports anonymous mode).	Optional; relies on web headers (Basic/Bearer tokens), cookies, or OAuth at application layer.